Your idea solves a **real problem**. Almost every college spends hours manually arranging students before exams. If you build this correctly, your website can be used by **VTU colleges, engineering colleges, universities, schools, and coaching institutes**.

🎯 Project Name

Smart Exam Seat Arrangement System (SESAS)

Tagline:

"Generate Fair Examination Seating in One Click"

🎯 Main Goal

Instead of manually arranging 1000-5000 students,

The admin should only

- Upload student list
- Upload classrooms
- Select seating rules
- Click **Generate Arrangement**

Within **5-10 seconds** the system generates

- Seat numbers
- Hall allocation
- Classroom layout
- Invigilator list
- Attendance sheets
- Student seat slips
- PDF & Excel reports

👥 Users

1. Super Admin

College principal

2. Exam Coordinator

Creates examination

3. Department Staff

Uploads students

4. Students

Search their seat

🏠 Website Flow

Home

↓

Login

↓

Dashboard

↓

Create Exam

↓

Upload Student Excel

↓

Upload Classroom Details

↓

Select Arrangement Rules

↓

Generate Seat Arrangement

↓

Preview

↓

Download PDF / Excel

↓

Student Search

Dashboard

College Name

Upcoming Exams

Today's Exams

Total Students

Total Rooms

Generate Seating

Student Search

Reports

Settings

Logout

Module 1

College Information

Store

- College Name
- Logo
- Address
- Semester
- Academic Year

Module 2

Student Database

Upload Excel

Example

USN	Name	Branch	Semester
1	Rahul	CSE	4
2	Kiran	ECE	4
3	Asha	ISE	4

Search

Edit

Delete

Export

Module 3

Classroom Database

Example

Room	Rows	Columns	Capacity
101	5	6	30
102	5	6	30
103	4	5	20

Also include

Bench Capacity

`1 student`

or

`2 students`

Module 4

Examination

Create

Example

Exam Name

Semester

Date

Time

Subject

Duration

🗂 Module 5

Seating Rules

This is the most important module.

Administrator can enable or disable rules.

Example

☑ Separate branches

☑ No same branch left-right

☑ No same branch front-back

☑ Alternate branch arrangement

☑ Maintain USN order

☑ Fill room completely

☑ Avoid empty benches

☑ Randomize seats

☑ One student per bench

☑ Two students per bench

Main Algorithm

Input

Students

```
CSE = 120

ISE = 90

ECE = 100

EEE = 80

ME = 50
```

Rooms

```
101 = 30

102 = 30

103 = 30

104 = 40

105 = 50
```

Step 1

Group students

```
CSE

ISE

ECE

EEE
```

ME

Step 2

Sort

Branch

Semester

USN

Step 3

Create Queue

Queue CSE

Queue ISE

Queue ECE

Queue EEE

Queue ME

Step 4

Round Robin

Take

`1 CSE`

↓

`1 ECE`

↓

`1 ISE`

↓

`1 EEE`

↓

`1 ME`

↓

`Again`

This ensures no branch dominates the seating order.

Step 5

Seat Validation

Before placing a student in seat `(r, c)`, check neighboring seats:

- Left
- Right (if already assigned)
- Front
- Back (if already assigned)

If any adjacent student belongs to the same branch, reject that placement and try the next student from another branch.

Step 6

Room Allocation

Fill

`Room 101`

↓

`Room 102`

↓

`Room 103`

until all students are seated.

Step 7

Conflict Resolution

If no branch satisfies the adjacency rules for a seat:

- Try swapping with previously placed students.
- If still impossible (e.g., one branch has far more students than others), relax the least important rule and log the exception.

Advanced Algorithm (Recommended)

Use a **Constraint Satisfaction Problem (CSP)** approach with **backtracking** and **forward checking**.

Treat:

- Each seat = variable
- Each student = value
- Rules = constraints

This produces a much more reliable arrangement than a simple greedy algorithm, especially for large datasets.

Generated Reports

1. Room Wise PDF

```
Room 101

Seat 1

Rahul

CSE

Seat 2

Kiran

ECE

...
```

2. Attendance Sheet(according to the branchwise)

USN

Name

Branch

Signature

3. Invigilator Report

Room

Faculty

Time

🔍 Student Portal

Student enters

USN

Output

Name

Room

Bench

Seat

Subject

Exam Time

📊 Admin Analytics

- Total students
- Total rooms
- Branch-wise count
- Room utilization
- Empty seats
- Rule violations (if any)
- Arrangement time
- PDF downloads

🔐 Security

- Role-based login (Admin, Coordinator, Staff)

- Audit logs
- Automatic backups
- Data encryption
- Password hashing

🛠 Suggested Technology Stack

Frontend

- Html 5
- CSS 5
- Java Script

Backend

- FastAPI (Python)

Database

- supabase

Excel Processing

- Pandas
- OpenPyXL

PDF Generation

- ReportLab